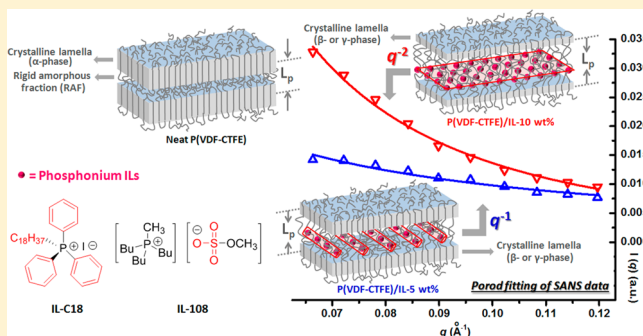


Understanding of Versatile and Tunable Nanostructuration of Ionic Liquids on Fluorinated Copolymer

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S Supporting Information

ABSTRACT: A fundamental understanding of the versatile and tunable nanostructuration effect of two phosphonium ionic liquids (ILs), denoted octadecyltriphenylphosphonium iodide (IL-C18) and tributyl(methyl)phosphonium methylsulfate (IL-108), on a semicrystalline fluorinated copolymer poly(vinylidene fluoride-co-chlorotrifluoroethylene) (P(VDF-CTFE)) was elaborated. This research suggests that the nanostructuration mechanism is very dependent on the diffusion and interacting abilities of ILs in matrix, which are determined by the chemical structures of ILs. Both ILs appear to diffuse and regularly assemble in the rigid amorphous fraction (RAF) of P(VDF-CTFE) between crystalline lamellae thanks to the “template” confinement effect of 2D structure. The assembled structure shows a gradually regular 1D to 2D geometric evolution with increasing the ILs amount revealed by Porod exponents in SANS analyses. Simultaneously, these regularly assembled ILs in the RAF also act as “template” to efficiently induce a complete transition of nonpolar α -phase to polar γ - and/or β -phase with only a tiny amount of ILs (from 2 wt %) (by XRD and FTIR). Moreover, the crystallization displays a depression behavior with reduced T_c , T_m , crystallinity (by DSC), and crystallite size (by SANS) due to the existence of ILs in the RAF seen as obstacle for the regular chain-folding and the dipolar interaction with P(VDF-CTFE). We propose that the synergistic influence of mutual “templates” endues phosphonium ILs a versatile nanostructuration on the fluorinated matrix. Moreover, this versatile nanostructuration effect can be easily tuned by tailoring the chemical structures of ILs. The difference in the chemical structures of ILs endows them different diffusion rates upon crystallization from solution for preparing resulting blend films and strengths of dipolar interaction with P(VDF-CTFE), consequently affecting their dispersion state in the matrix to reside in the RAF with different extents and interacting abilities (confirmed by DEA and DMA). Thus, IL-108 with faster migration rate and stronger dipolar interaction than IL-C18 (due to the smaller steric hindrance and extra polar groups in the cation/anion combination of the IL structure) locates in the RAF much more to induce more *trans*-configuration (β -phase), more homogeneous dispersion morphology, and higher depression of crystallization.



1. INTRODUCTION

In the development of advanced polymer materials with significantly unprecedented and improved physicochemical properties, the concept of “functionalization via structural design” is a critical guideline to tailor the matter structuration at nanoscale.¹ Since the first use of a small amount of ionic liquids (ILs) in polytetrafluoroethylene (PTFE) matrix to achieve a tunable nanostructuration on the morphologies and mechanical properties was innovatively discovered in our group,^{2,3} increasing interest has been developed to prepare functional polymer materials with tailored nanostructures by the aid of ILs as nanostructuration agents. The attractiveness of ILs arises from their simple synthetic route and low-content requirement over the mostly used amphiphilic block copolymers,^{4–7} silica,⁸ layered silicates,^{9,10} carbon nanotubes (CNTs),^{11,12} etc., for many years. For example, imidazolium^{13,14} and phosphonium^{14–18} ILs were used as functional additives to achieve diglycidyl ether of bisphenol A (DGEBA)-based epoxy

networked polymeric materials which display a structuration at nanoscale with dramatic mechanical properties. More recently, phosphonium ILs were used as nanostructuration agents to develop a biopolymer matrix poly(butylene adipate-co-terephthalate) (PBAT) with significantly improved water vapor permeability and mechanical properties.¹⁹ Therefore, a virgin application of ILs as efficient nanostructuration agents has been exploited to achieve nanostructured polymeric materials with tunable properties, more frequently used as “green” solvents for organic synthesis, catalysis, fuel cells, electrochemistry, etc.²⁰

In terms of one of the most important electroactive fluoropolymers, poly(vinylidene fluoride) (PVDF) and its co(ter)polymers have earned a large range of applications due

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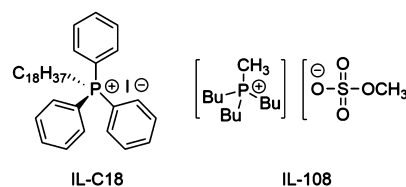
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to pyro/ferro/piezo properties.²¹ Among the most frequent α , β , and γ crystalline phases, polar β - and γ -ones present superior electrical properties than nonpolar α -one due to the relative large spontaneous polarization per unit cell. The electroactive performance of PVDF-based materials is found to be tightly associated with the nanostructure; thus, the tunable nanostructuration on PVDF-based matrices is a key issue for versatile applications. Recently, few works report the effect of imidazolium ILs on the structures and properties of PVDF matrix. Wang et al.²² prepared β -phase-dominant PVDF/IL (1-ethyl-3-methylimidazolium nitrate) thin films spin-coated from a solution. IL promotes the formation of ferroelectric β -phase and much smaller crystalline regions. The ferroelectric properties show a remanent polarization of 60 mC/m² and a quasi-static pyroelectric coefficient of 19 $\mu\text{C}/(\text{m}^2 \text{ K})$ at 30 °C. Xing et al.²³ systematically investigated the crystalline structures, miscibility, and physical properties of PVDF/IL (1-butyl-3-methylimidazolium hexafluorophosphate) blend films. It was found that the IL displays drastic effects on both structure and properties of PVDF, generating a multifunctional film. Moreover, by the aid of electrospinning technique, the resulting nanofibers of PVDF/IL blends exhibiting extremely high β -phase content were obtained.²⁴ The synergistic effect of ILs combined with CNTs influencing the structure of PVDF crystalline regions has also been illustrated.^{25,26} The resulting materials exhibit improved mechanical and electronic conductivity properties due to the anchored ILs on the surface of CNTs. The latest case is that PVDF nanoparticles with α -phase were partly transformed into β -one by adding imidazolium nitrate IL and subsequent thermal annealing below the melting temperature.²⁷ Although the influence of IL on the structure of PVDF (mainly focusing on the formation of polar crystalline phase) has been observed above, the mechanism for the structuration is quite elusive. From current understanding, dipolar interactions between PVDF chain molecules and ILs contribute the formation of polar crystalline phase. Moreover, in order to obtain more β -crystals, the long tangible template, IL-covalently modified CNTs, was suggested to be the second factor to provide long enough sequences and stabilize the high-energy of *all-trans* conformation of PVDF. In essence, understanding of the fundamental mechanism of the versatile nanostructuration of ILs on semicrystalline PVDF-based polymer is still superficial, such as (i) what determines the strength of dipolar interaction, (ii) how does the dipolar interaction affect the nanostructuration on the matrix, not only the crystalline phase structure but also the dispersion morphology and crystallization behavior, and (iii) what is the relationship between the nanostructuration effect of ILs and their localization in matrix and also what governs this localization?

In view of advantages of phosphonium ILs over imidazolium and pyridinium counterparts which are widely used in polymer/IL blends, they (i) are thermally more stable, which is desirable for processing at greater than 200–300 °C; (ii) have no acidic proton, which avoids the formation of carbene; and (iii) exhibit higher electrochemical stability, which has been used as electrolytes with excellent electrochemical performance.^{28,29} In this work, two phosphonium ILs, denoted octadecyltriphenylphosphonium iodide (IL-C18) and tributyl(methyl)-phosphonium methylsulfate (IL-108) (Scheme 1), are incorporated in a PVDF copolymer poly(vinylidene fluoride-*co*-chlorotrifluoroethylene) (P(VDF-CTFE)) to prepare two series of blend films with different IL amounts via solution

Scheme 1. Molecular Structures of Two Phosphonium ILs Used in This Paper



casting from methyl ethyl ketone (MEK) solution. The differences in steric hindrance and extra dipolar groups are considered as potential key factors to influence the interaction with P(VDF-CTFE) and diffusion rate of ILs in polymer matrix during the film formation. The crystalline phase structure, dispersion morphology, and crystallization behavior are finely characterized with the goal of providing a full and deep understanding of the versatile and tunable nanostructuration effect of phosphonium ILs. We propose that the essence of structuring mechanism is closely related to the chemical structure of IL. Thus, such a fundamental understanding of the nanostructuration of ILs will help to design and tune this unique class of electroactive PVDF-based materials by the aid of ILs for advanced applications.

2. EXPERIMENTAL SECTION

Materials and Blend Films Preparation. P(VDF-CTFE) copolymer containing 8 wt % of CTFE was purchased from Arkema. IL-108 is commercially available CYPHOS-IL108 from Cytec (USA). The synthesis of IL-C18 was described elsewhere.² The solvent methyl ethyl ketone (MEK) (chemically pure) was purchased from CARLO ERBA (France) and used as received.

Twelve wt % of P(VDF-CTFE) solution in MEK was first prepared at 60 °C. Different amounts of ILs (IL-C18 or IL-108) were added into the solution to generate different compositions; i.e., the ratio of IL over P(VDF-CTFE) was 2, 5, and 10 wt %. After adequately stirring at 60 °C, the films were solution-casted from the mixture on a stainless steel substrate via doctor blade, followed by drying at 80 °C in an air-circulation oven for 3 h to remove any residual solvent. The obtained films (around 15–30 μm) were peeled from the substrate and marked as P(VDF-CTFE)/IL-C18 or P(VDF-CTFE)/IL-108 for different compositions.

Characterizations. The X-ray diffraction (XRD) data were collected using a Bruker D8 Advance X-ray diffractometer. A bent quartz monochromator (before the sample) was used to select the Cu K α radiation ($\lambda = 0.154\,06\text{ nm}$) and run under operating conditions of 45 mA and 33 kV in a Bragg–Brentano geometry. The angle range scanned is $1^\circ < 2\theta < 30^\circ$ at a scanning speed of $2^\circ/\text{min}$ with a step interval of 0.02° .

Fourier transform infrared (FTIR) spectra were recorded at room temperature on a Nicolet ISO10 Thermo Scientific spectrometer. Note that the resulting films under the conditions we used are too thick to obtain perfect absorption peaks in the characteristic range of 1100–1500 cm^{-1} (some flat peaks are visible, data not shown) if only transmission mode is used. Thus, both transmission and attenuated total reflectance (ATR) modes (a diamond crystal was used) were performed to determine each characteristic band. All spectra were recorded at 2 cm^{-1} resolution and 32 scans from 4000 to 400 cm^{-1} for transmission mode and 4000 to 650 cm^{-1} for ATR mode. Although two modes were used, all characteristic absorption peaks are in agreement with each other.

Differential scanning calorimetry (DSC) analyses were conducted on a thermoanalyzer system, model Q20 (TA Co. Ltd., USA), under a nitrogen atmosphere at a rate of 10 °C/min. Four scans (heating/cooling/heating/cooling, separated by a 10 min isotherm) were realized for each sample between -70 and 180 °C. All nonisothermal crystallization temperatures (T_c) from the first cooling run and melting

temperatures (T_m) from the second heating run reported here correspond to the summit point of each peak. The crystallinity (χ_c) is obtained from the enthalpy change in the second heating run. The values are calculated by considering $\Delta H_{100\%}$ (melting enthalpy for totally crystalline PVDF) to be 104.9 J/g.³⁰

Transmission electron microscopy (TEM) was performed at the Center of Microstructures (Université de Lyon) using a Philips CM 120 field emission scanning electron microscope with an accelerating voltage of 80 kV. The samples were cut using an ultramicrotome equipped with a diamond knife to obtain 60 nm thick ultrathin sections. Then, the stained sections by RuO₄ were set on copper grids for observation.

Small-angle neutron scattering (SANS) experiments were carried on the small-angle spectrometer PAXY spectrometer of Léon Brillouin Laboratory (Saclay, France). Three sample-to-detector distances (SDD) and neutron wavelengths (λ) were used to cover magnitudes of the scattering vector modulus q from 0.03 to 4.2 nm⁻¹ ($\lambda = 1.2$ nm, SDD = 6.75 m, $\lambda = 1.2$ nm, SDD = 3 m, and $\lambda = 0.5$ nm, SDD = 1 m). The resulting level of the scattering intensity at the high q values is of the order of magnitude of the experimental background, notably the incoherent contributions of the polymer phase and ionic liquids are subtracted.

Dynamic mechanical analysis (DMA) was investigated by a Rheometrics Solid Analyzer RSA II at 0.01 wt % tensile strain and a frequency of 10 Hz. The heating rate was 3 K/min for the temperature range from -80 to 120 °C under a nitrogen atmosphere.

Dielectric analysis (DEA) measurements were carried out using a Novocontrol Alpha-A analyzer (BDC Novocontrol). Samples for DEA tests were cut out from the films as 25 × 25 mm². Gold electrodes of circular size with diameter 10 mm were deposited by sputtering. The applied voltage is 1 V RMS. The temperature of the sample was controlled with a stability of $\Delta T = 0.1$ °C (Novocontrol Quatro system controller BDS 1330). Two cycles were carried out for the frequency domain 10⁻¹–10⁶ Hz and for temperature from -80 to 100 °C in a step of 2 °C/min.

Uniaxial tensile tests were carried out using a MTS 2/M electromechanical testing machine at 22 ± 1 °C and 30 ± 5% relative humidity at a crosshead speed of 5 mm/min. The strain was calculated from the crosshead displacement. The tensile tests were performed at least three samples. A cookie cutter was used to obtain dumbbell-shaped specimens.

3. RESULTS AND DISCUSSION

Identification of Crystalline Phase. The XRD spectrum of neat P(VDF-CTFE) in Figure 1 unambiguously presented three diffraction peaks at $2\theta = 17.9^\circ$, 19.9° , and 27.2° , which shows remarkable similarities with PVDF corresponding to the α -crystal planes with Miller indices (100), (110), and (021), respectively. The P(VDF-CTFE) copolymers containing small content of CTFE (<30 mol %) give rise to a monoclinic crystalline structure in which the comonomer CTFE is not able to accommodate within the VDF unit cell, i.e., not cocrystallizable with VDF units but in the amorphous phase.²¹ Concerning the low content of CTFE (8 wt %, i.e., 4.6 mol %) in this paper, the crystalline nature of P(VDF-CTFE) copolymer is due to the PVDF-like crystal, exhibiting the same 2θ values as PVDF homopolymer and relatively broad diffraction peaks. The corresponding FTIR spectra in both ATR mode (in the range of 1600–650 cm⁻¹) (Figure 2A) and transmission (in the range of 700–400 cm⁻¹) mode (Figure 2B) also indicate only all characteristic absorption peaks of α -phase at 408, 532, 615, 761, 796, 854, 974, and 1383 cm⁻¹,^{23,31,32} suggesting that the neat P(VDF-CTFE) exclusively exists in the α -phase which is the most common polymorph produced from the solution casting. As IL-C18 is added in P(VDF-CTFE) with the quantity from 2 to 10 wt %, the characteristic diffraction peaks of α -phase were not traced, but

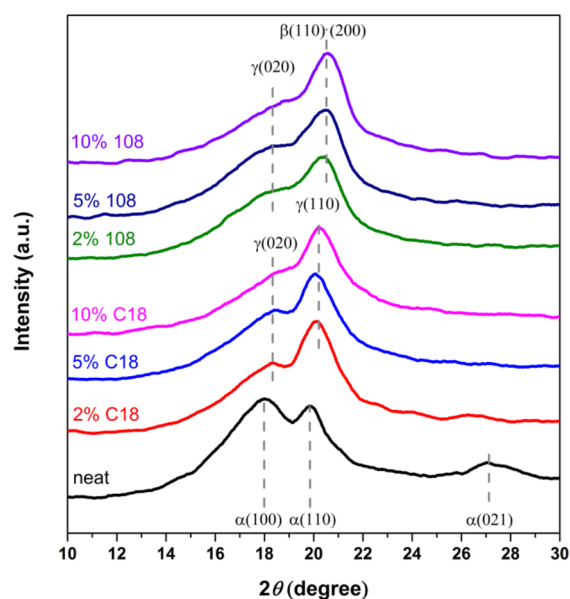


Figure 1. XRD patterns of neat P(VDF-CTFE) and P(VDF-CTFE)/ILs blends.

two new peaks are observed at $2\theta = 18.5^\circ$ and 20.1° , corresponding to $\gamma(020)$ and $\gamma(110)$ crystal planes. The FTIR spectra also show the complete removal of α -phase and the obvious appearance of new bands of γ -phase at 430, 510, 812, 834, and 1234 cm⁻¹, confirming almost entire transformation of α -phase to γ -phase. With adding IL-108 in P(VDF-CTFE), the diffraction peaks uniquely assigned to α -phase are not observed, but only two major peaks at $2\theta = 18.5^\circ$ and 20.5° are measured for all blends regardless of the amount of IL, assigned to $\gamma(020)$ and $\beta(110)(200)$ planes, respectively.^{33,34} The corresponding FTIR spectra display no characteristic bands of α -phase but if γ -phase at 430, 510, 812, 1234 cm⁻¹ and new typical bands of β -phase at 445 and 1274 cm⁻¹ as well as band at 840 cm⁻¹ ascribed to β - or γ -phase,^{23,31} suggesting that all nonpolar α -phase is converted to a mixture of polar β - and γ -phase. Note that the introduction of the bulky comonomer CTFE, as the crystalline defect into the PVDF backbone, actually imposes a large steric hindrance and makes polymer chains loose, which decreases the energy barrier of phase transformation and thus facilitates the *trans*-conformation of the polymer backbone.^{35,36} However, a theoretical work by Ranjan et al.³⁷ indicates that for those P(VDF-CTFE) containing CTFE lower than 17 mol %, the nonpolar α -phase is the more thermodynamically stable form (with lower energy) than polar β -phase. With increasing the CTFE content in the copolymer, this difference in the energy between α -phase and β -phase decreases, so the transformation to β -phase becomes progressively easier until a critical value is approached (17 mol %) at which the energy difference decreases down to zero. Above 17 mol % of CTFE in the copolymer, the β -phase is preferred and can be obtained easily. Because of the low content of CTFE in our case, the effect of CTFE on the formation of polar phases is negligible, and that is why only α -phase is obtained in the neat P(VDF-CTFE). Thus, the obvious phase transformation behavior from nonpolar α -phase to β - and/or γ -phase is only due to the incorporation of phosphonium ILs.

Combining XRD and FTIR results, it is significant to show that a tiny amount of both phosphonium ILs can transform the

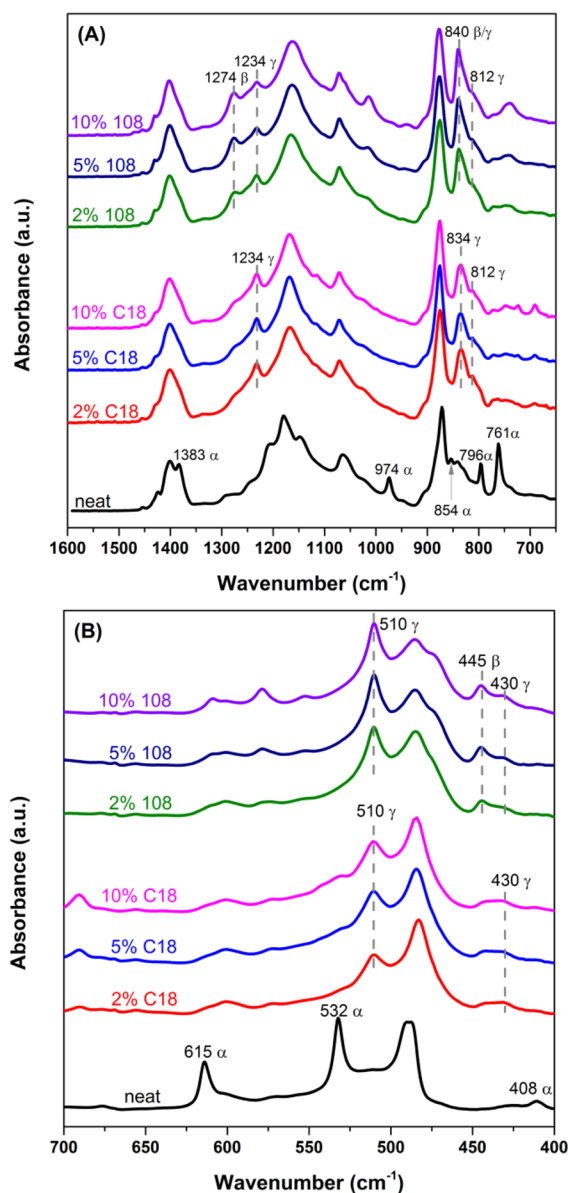


Figure 2. FTIR spectra of neat P(VDF-CTFE) and P(VDF-CTFE)/ILs blends in ATR mode (A) and transmission mode (B).

nonpolar α -phase to polar phases very efficiently (only 2 wt % of ILs is enough to induce 100% of crystalline phase transformation), and IL-108 can even induce some *all-trans* β -phase with longer TTTT sequence than that γ -phase with shorter TTT sequence. It is therefore an easy and efficient method to prepare polar phase films with large spontaneous polarization profile applying to a wide range of electroactive polymeric materials fields. This use of phosphonium ILs is so far superior to recent numerous endeavors such as incorporating pristine CNTs,³⁴ IL-modified CNTs,^{25,26} poly(methyl methacrylate) (PMMA) functionalized multiwalled carbon nanotubes (MWNT),³⁸ ester functionalized MWNT,³⁹ organically modified montmorillonite (OMMT),⁴⁰ mesoporous SiO₂ nanorods,⁴¹ and imidazolium ILs combining electrospinning technique²⁴ with the aid of electrical poling and uniaxial stretching, etc., avoiding using expensive nanofillers, complicated chemical modification, and the sophisticated electrospinning instruments.

Dispersion Morphology: Nanostructuration Effect.

The dispersion of ILs in P(VDF-CTFE) matrix can be well reflected by signals of SANS which are sensitive to the different nuclei density fluctuation among different phases in the semicrystalline matrix. Figure 3 shows the SANS profiles

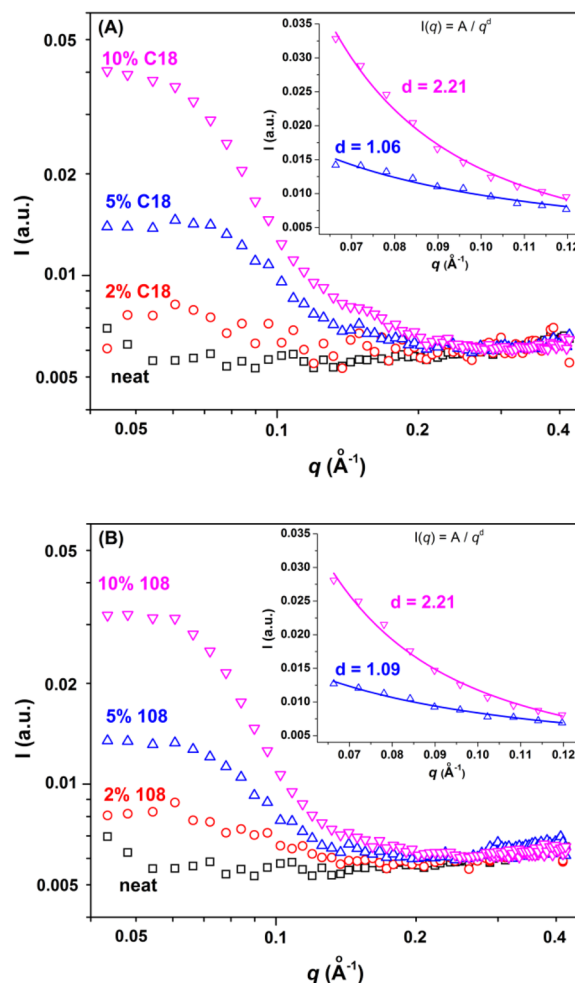


Figure 3. SANS patterns of neat P(VDF-CTFE) and blends with IL-C18 (A) and IL-108 (B). Insets display the Porod fitting in the modest q range.

focusing on the modest- q regime of films with different IL incorporations. It is well-known that the crystalline phase has higher nuclei density while the amorphous phase exhibits lower one. The scattering intensity is proportional to the density difference between them. However, very weak and random nuclei density fluctuations exist in neat P(VDF-CTFE) and those blends containing 2 wt % of ILs, potentially due to the limited contrast between crystalline and amorphous phase of P(VDF-CTFE), and that 2 wt % is not sufficient to enhance the contrast, respectively.⁴² We thus can consider them as a “homogeneous” system. However, the scattering intensity is subjected to increase when more ILs (from 5 to 10 wt %) incorporated in P(VDF-CTFE) matrix. We thus can conclude that the nuclei density difference has increased large enough to be seen as a “heterogeneous” system due to the incorporation of phosphonium ILs in matrix. Moreover, these Porod regimes in the range of $0.07 \text{ \AA}^{-1} < q < 0.12 \text{ \AA}^{-1}$ for blends with 5 and 10 wt % ILs are fitted with $I(q) = A/q^d$, represented by solid lines in the insets of Figure 3, where d is the Porod exponent yielding

information about the so-called “fractal dimension”, i.e., estimation of the geometric form of scattering objects.^{43,44} For P(VDF-CTFE)/ILs containing 5 wt % of ILs, d values stay around 1, which indicates the formation of 1D rod-like scattering objects, whereas they increase up to around 2 when 10 wt % of ILs are incorporated, reflecting the clearly detected 2D scattering objects like lamellae. The regular geometric scattering objects as well as the evolution from 1D to 2D profile with increasing IL amount indicate the regular assembly of ILs in semicrystalline matrix. It has been accepted widely that a semicrystalline polymer can be considered as a heterogeneous mixture consisting of crystalline phase and amorphous phase. The disordered chains in the amorphous phase present different mobility depending on the location in regard with crystalline phase, i.e., the rigid amorphous fraction (RAF) confined between crystalline lamellae and the mobile amorphous fraction (MAF) with fully “free” disordered chain conformation. RAF is an intermediate nanophase existing at the interface between crystalline phase and MAF as a result of the immobilization of a polymer chain due to the covalently coupling of crystals occurring across the crystalline phase boundaries.⁴⁵ First, due to the long-range disordered isotropic nature of MAF, the dispersion of ILs in the MAF cannot induce the observed regular scattering objects. Second, XRD spectra show no change of crystalline diffraction peaks with increasing the amount of ILs, indicating no dispersion of ILs in crystalline phase cocrystallized with P(VDF-CTFE). Thus, the most reasonable explanation to the evolution of 1D to 2D scattering objects is the regularly assembled structure of ILs in the RAF of matrix. These detected regularly geometric scattering objects by nuclei density fluctuation are actually seen as the low-dimensional arrangements of ILs assembled in the confined RAF between crystallites. Therefore, the 2D lamellar crystallite of P(VDF-CTFE) permits a so-called “template” confinement effect on the diffusion and dispersion of ILs in the RAF. The localization of ILs in the RAF will be further confirmed in the following microstructural characterization section.

The dispersion morphologies of ILs in the matrix are visually evaluated by TEM in Figure 4. Two types of phosphonium ILs

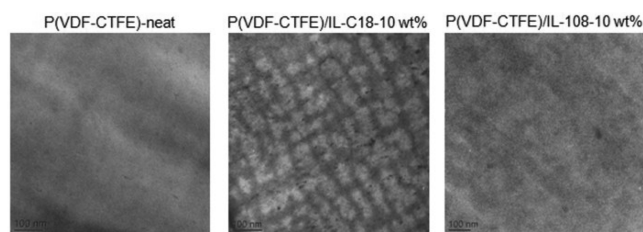


Figure 4. TEM images of P(VDF-CTFE)-neat and P(VDF-CTFE)/ILs-10 wt % blends.

display different dispersion degrees at nanoscale. P(VDF-CTFE)/IL-108 presents a homogeneous and cocontinuous morphology of highly dispersed ILs in matrix without aggregates at nanoscale revealed by RuO₄ marking, which is similar to the neat copolymer. For P(VDF-CTFE)/IL-C18, the image obtained by RuO₄ staining for locating phenyl groups in IL-C18 obviously shows enriched IL-C18 aggregation displayed by black domains with well-ranged grid-like morphology at relatively larger scale. The origin of different dispersion morphologies is likely due to the interaction between them.

Interaction between P(VDF-CTFE) and Phosphonium ILs. In view of molecular level, we further investigate the interactions between P(VDF-CTFE) and two types of phosphonium ILs. The dipolar interaction has been identified as Coulomb attraction between dipoles in PVDF and ions in IL.^{22,23} Therefore, the driving force governing the nano-structuration effect of ILs on P(VDF-CTFE) matrix is the dipolar interaction between dipoles in P(VDF-CTFE) (C–F and C–Cl) and ions in phosphonium ILs. For both phosphonium ILs, the dipolar interaction can be confirmed by FTIR spectra shown in Figure 5. The amorphous peak of

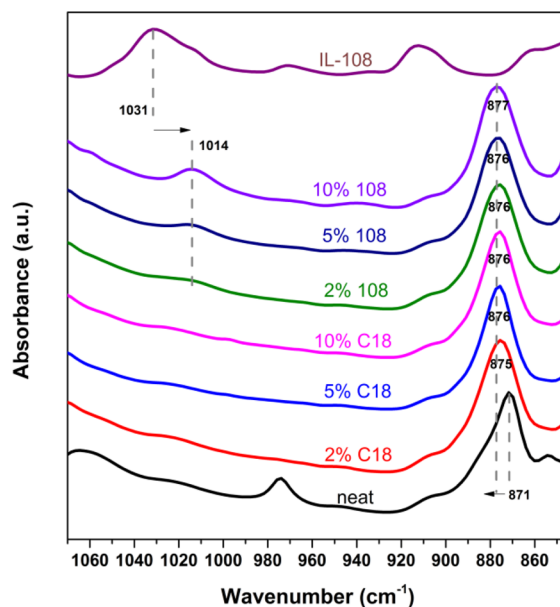


Figure 5. ATR FTIR spectra of neat P(VDF-CTFE), IL-108, and P(VDF-CTFE)/ILs blends.

–CF₂–CH₂– bending vibration in neat copolymer is located at 871 cm^{−1},^{23,39} and it is independent of the crystalline phase transformation.⁴⁶ This peak shows a blue-shift (moving to higher wavenumber) in all blends, and the peak positions for P(VDF-CTFE)/IL-C18-10 wt % and P(VDF-CTFE)/IL-108-10 wt % are at 876 and 877 cm^{−1}, respectively. The blue-shift could be ascribed to a higher energy required for the –CF₂–CH₂– bending vibration because the dipoles in polymer matrix (C–F) are involved in the dipolar interaction with cations of phosphonium ILs. Note that IL-C18 has a bigger volume of phosphonium cation with three rigid phenyls and a long C18 alkyl chain substituent than that of IL-108, which to some extent hinders the accessibility between dipoles and thus dilute such dipolar interaction. Moreover, unlike IL-C18, IL-108 has extra S=O dipoles in its anion, inducing the additional dipolar interaction with those dipoles in copolymer matrix. As shown in Figure 5, IL-108 displays an absorption band at 1031 cm^{−1} attributed to symmetric –SO₃ stretching vibrations.⁴⁷ It is clear to see a red-shift (moving to lower wavenumber) to 1014 cm^{−1} for all blends containing IL-108. This red-shift reflects lower vibration energy needed for S=O dipole when it is involved in the dipolar interactions with P(VDF-CTFE) matrix because the electrostatic interaction of CH₃OSO₃[−] with phosphonium cation in the IL-108 was to some extent weakened and replaced by the intermolecular dipolar interaction of the phosphonium cation with P(VDF-CTFE) dipoles with lower energy requirement for bending vibration. Thus, in view of the

chemical structures of two phosphonium ILs used, we propose that the origin of the difference in the interaction strength between P(VDF-CTFE) and ILs (IL-108 > IL-C18) is twofold: (i) accessibility between ions in ILs and dipoles in P(VDF-CTFE); (ii) existence of more extra dipolar groups in the structure of ILs.

Concerning the different interaction strengths of P(VDF-CTFE) with IL-C18 and IL-108, it thus well explains why IL-108 can even induce some β -phase comparing with IL-C18: the stronger interaction capability of IL-108 with P(VDF-CTFE) can finely stabilize the TTTT conformation requiring higher energy,³⁴ consequently engendering more favorable β -phase. In addition, the stronger interaction also promotes more homogeneously dispersion of IL-108 in the P(VDF-CTFE) matrix without aggregates at nanoscale comparing with that of IL-C18 with larger nanoaggregates shown in TEM images (Figure 4). Therefore, the strength of affinity between phosphonium ILs and P(VDF-CTFE) matrix due to the chemical nature of ILs determines the resulting crystalline phase structure and dispersion morphology and should have a significant effect on the crystallization behavior of semicrystalline matrix with respect of the polymer high-ordered crystalline structure generated in the growth of crystalline lamellae.

Crystallization Behavior. T_c , T_m , and χ_c of P(VDF-CTFE)/ILs blends as a function of the IL amount are shown in Figure 6 (all DSC traces are shown in Figure S1 of Supporting

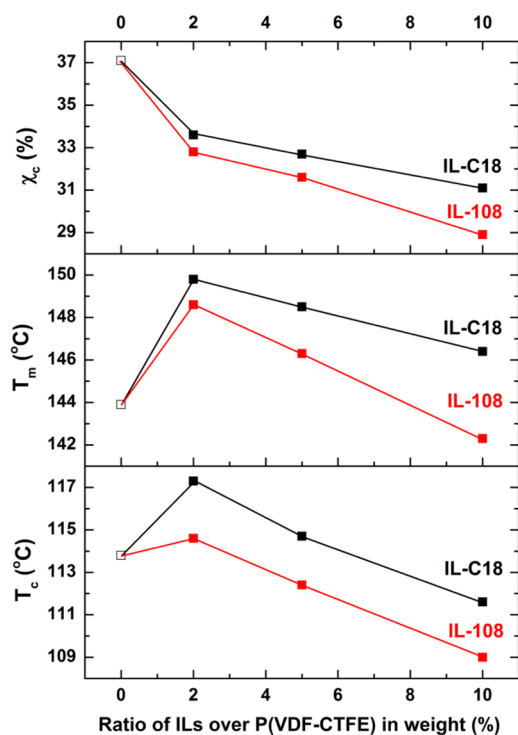


Figure 6. T_c , T_m , and χ_c as a function of ILs amount in P(VDF-CTFE)/ILs blends.

Information). Note that the blend film containing 2 wt % of either IL-108 or IL-C18 displays an abrupt enhancement in T_c and T_m compared with those of neat P(VDF-CTFE). The T_c increment behavior is due to the nucleation effect of ILs to accelerate the crystallization of polar γ - and/or β -phase which exhibit 6–7 °C higher T_m than nonpolar α -phase,^{48,49} subsequently leading to T_m increment in the second heating

run. For a given IL, one can also obviously see the T_c , T_m , and χ_c depressions as increasing the amount of ILs in the blends. For a given amount of IL, it is evident to see the different potentials to reduce T_c , T_m , and χ_c (IL-108 > IL-C18).

The depression effect of phosphonium ILs on crystallization behavior from the solution for preparing blend films can be further explored by the evolution of the size of lamellar crystallites calculated by using Debye–Bueche equation fitting with SANS data:⁴²

$$I(q) = \frac{I_0}{(1 + q^2\xi^2)^2}$$

where ξ is the correlation length related to the crystallite size and I_0 is the zero-angle scattering intensity. The low- q regime is best fitted, represented by solid line in Figure 7, and the calculated values of ξ are compared in Table 1. The same depression behavior of crystallite size can be illustrated by the ξ values. For neat P(VDF-CTFE), the ξ value is 159.2 Å, while the incorporation of both phosphonium ILs decreased the ξ value. Moreover, it is striking to see a more dramatic decrease of crystallite size by IL-108 than that by IL-C18 with the same IL amount (e.g., ξ = 130.1 Å for 10 wt % of IL-C18 versus ξ = 97.5 Å for 10 wt % of IL-108). Accompanying the decrease of lamellar crystallite size, the distance between them (i.e., interlamellar spacing) should also increase, which can be estimated by the average long period (L_p) defined as the center-to-center distance of the lamellar crystallites equaling to sum of the thickness of the lamellar crystallite and the rigid amorphous domain between them.^{42,50} The values of L_p (shown in Table 1) are calculated by using Bragg's law, $L_p = 2\pi/q_{\max}$ where $q_{\max}$ is the peak value found in the Lorentz-corrected plot (Iq^2 vs q) in Figure 8. However, one cannot observe the L_p peaks for neat copolymer and blends containing 2 wt % of ILs because of the limited contrast as mentioned above. The opposite variations of L_p and ξ values are observed for all blends with 5 and 10 wt % of ILs: decreasing the crystallites size leads to increasing the distance between them.

It is well-known that the crystallization behavior of the growth of lamellar structure for a semicrystalline polymer/filler blend actually is reassigning fillers in the matrix and consequently perturbing the dispersion of fillers from either solution or melt. For our P(VDF-CTFE)/ILs blends, this so-called redispersion of phosphonium ILs upon crystallization either from solution (SANS analyses based on as-prepared films) or melt (DSC analyses after the first heating run up to above melting temperature) is necessarily related to two factors: (i) the diffusion ability of ILs in the matrix; (ii) the interaction strength between them, which are all governed by the chemical structures of ILs. First, for both phosphonium ILs, we have shown no coexistence of ILs and P(VDF-CTFE) in the crystalline phase, so the most reasonable explanation to the effect of depressing crystalline phase is that ILs are able to migrate in the RAF confined between lamellar crystallites to affect the chain folding, which agrees with the Porod fitting results in SANS analyses. Those ILs existing in the RAF definitely act as obstacles for the formation of crystallites and thus reduce the crystal growth rate, exhibiting a lower T_c in the nonisothermal crystallization and then producing a relatively imperfect crystalline phase comparing with neat P(VDF-CTFE) with a lower T_m , reduced χ_c , and crystallite size. Second, the affinity of both ILs with P(VDF-CTFE) which has been confirmed above restrains the process of regularly folding

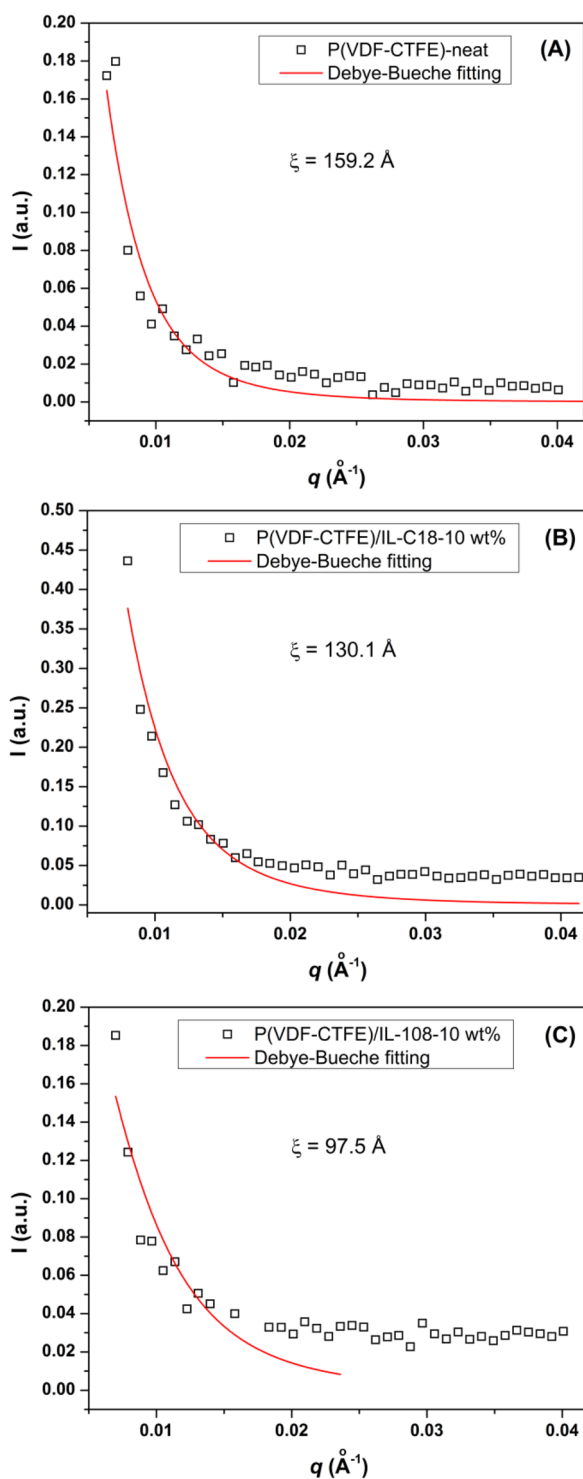


Figure 7. Debye–Bueche fitting of neat P(VDF-CTFE) (A) and blends with 10 wt % of IL-C18 (B) and IL-108 (C).

Table 1. Summary of SANS Analyses

sample	correlation length (ξ) ($\text{\AA}$)	long period (L_p) ($\text{\AA}$)
P(VDF-CTFE)-neat	159.2 ± 15.2	
P(VDF-CTFE)/IL-C18-5 wt %	133.6 ± 17.0	65.6
P(VDF-CTFE)/IL-C18-10 wt %	130.1 ± 18.4	80.5
P(VDF-CTFE)/IL-108-5 wt %	100.3 ± 13.2	74.6
P(VDF-CTFE)/IL-108-10 wt %	97.5 ± 15.4	87.1

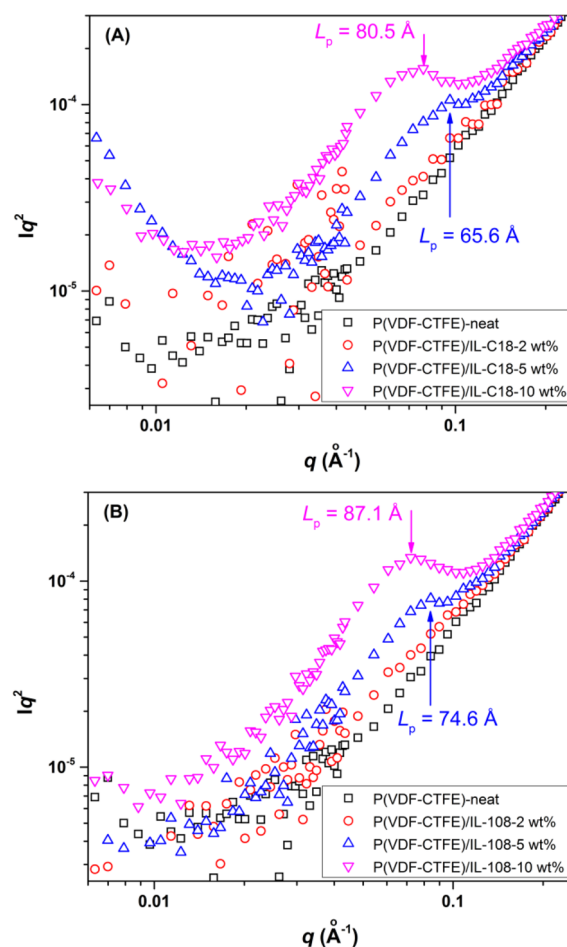


Figure 8. Lorentz-corrected scattering patterns of neat P(VDF-CTFE) and blends with IL-C18 (A) and IL-108 (B).

chain to form crystalline phase because any chain segment movement to change the configuration is restricted by neighboring ILs due to the interaction between them. In addition, the different degrees of depression on the crystallization behavior of P(VDF-CTFE) matrix by using two types of phosphonium ILs (IL-108 > IL-C18) are thus due to the difference in the ability of diffusion motion of ILs and in the interaction strength when blend films are formed from the solution containing polymer, ILs, and solvent. In view of the chemical structure, three benzenic rings and a long C18-alkyl chain of IL-C18 cause bulky spatial distribution, and that makes difficulty in diffusion motion in the matrix. Oppositely, IL-108 with smaller volume than IL-C18 diffuses faster to reside in the RAF in greater magnitude, consequently depressing the crystallization much more. Combining to the stronger dipolar interaction of IL-108 with matrix than that of IL-C18 due to the chemical structure, the depression effect on crystallization behavior is thus very dependent on the chemical structures of ILs.

Microstructural Characterization: Localization of ILs.

The versatile and tunable nanostructuration effect of phosphonium ILs on crystalline phase structure, dispersion morphology, and crystallization behavior of P(VDF-CTFE) matrix has been in-depth investigated. We suggested that the localization of ILs in the RAF between lamellar crystallites contributes the structuring actions, so the further insight into the microstructure to understand the localization of ILs in the

matrix is not only a direct proof to interpret the IL structuring mechanism but also important to forecast electroactive properties dependent on the tunable microstructures. The dynamics of amorphous segmental relaxation in both non-confined and confined environments are the most effective means to identify the MAF and RAF of amorphous phase in semicrystalline polymers such as poly(phenylene sulfide),⁵¹ poly(ether ether ketone),⁵² polycarbonate,⁵³ and polylactides.⁵⁴

Figure 9 shows the DEA curves for dielectric loss tangent ($\tan \delta$) and the imaginary part of dielectric modulus (M'') as a

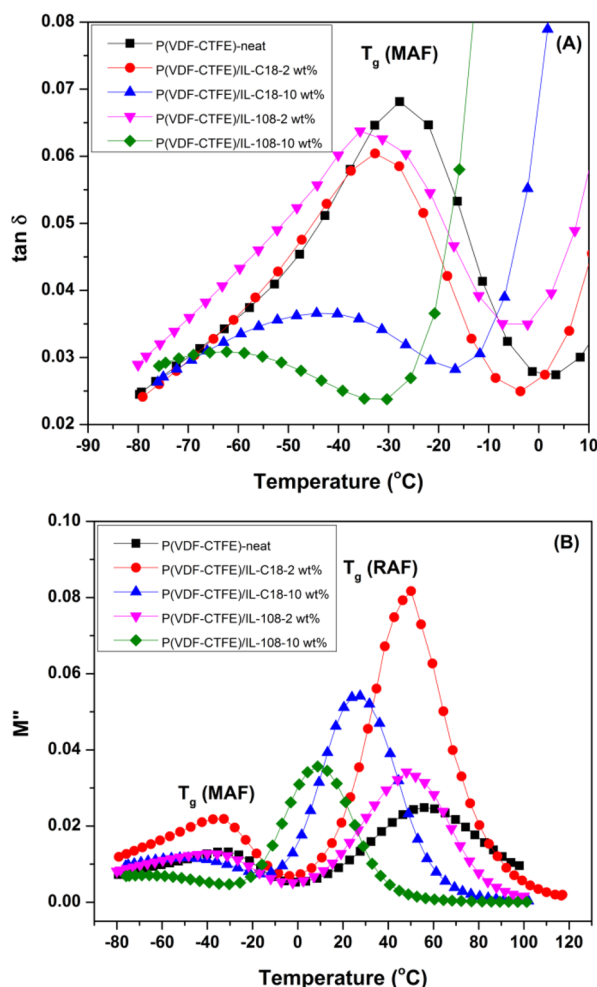


Figure 9. Dielectric loss tangent ($\tan \delta$) (A) and the imaginary part of the dielectric modulus (M'') (B) as a function of temperature of neat P(VDF-CTFE) and blends with ILs ($f = 1$ kHz, the second heating run).

function of temperature. In Figure 9A, a single dissipation peak centered at -28 °C of neat matrix corresponds the glass transition temperature (T_g) of MAF of P(VDF-CTFE). For a given IL, the T_g (MAF) decreases with increasing the ILs amount. For a given amount of IL, blends with IL-108 display a lower value of T_g (MAF) than IL-C18 (see Table 2 for each T_g value). The change of T_g (MAF) verifies the localization of both phosphonium ILs in the MAF through dipolar interaction with P(VDF-CTFE) chains. On the one hand, the interaction of ILs with polymer chains by incorporating small molecular ILs in the MAF of P(VDF-CTFE) weakens the original interchain affinity to enhance the motion freedom with increased free volume. On the other hand, the already confirmed effect of ILs on the depression of crystalline phase gives rise to a lower crystallinity and smaller crystallite size and thus makes more amorphous room for the free volume. Combining these two factors, the increment of free volume in the MAF contributes to the T_g depression. Moreover, the stronger interaction with matrix and depression behaviors on crystallinity and crystallite size allow IL-108 to induce lower T_g s than IL-C18 with the same IL amounts. The T_g depression behavior actually indicates the plasticizer effect of ILs on the matrix, which can be further confirmed by the macroscopic mechanical property via the strain–stress tests (see Table S2 in Supporting Information). Comparing the results of neat P(VDF-CTFE) and blends containing 10 wt % of ILs, it is noticeable that the incorporation of IL reduced the Young's modulus, suppressed the yield strength, and improved the elongation at break. It is also significant to find a dramatic enhancement of elongation at breakup to 838% for P(VDF-CTFE)/IL-108-10 wt % film, whereas only 122% for the P(VDF-CTFE)/IL-C18-10 wt % one. In Figure 9B, besides the T_g (MAF) relaxation with the same depression behavior, other relaxation peaks centering at relatively high temperature are visible. These relaxations are due to T_g in the RAF confined between crystalline lamellae because a higher temperature is required to acquire sufficient energy to mobilize these so-called constrained chains in the RAF.^{54,55} Incorporating both phosphonium ILs also leads to the same T_g (RAF) variations as T_g (MAF) due to the increment of free volume in the RAF (see Table 2 for each T_g value).

These two relaxations occurring in the MAF and RAF can also be seen in DMA curves in Figure 10. All temperature values of relaxation peaks are summarized in Table 2. A loss $\tan \delta$ peak centered at -45 °C for neat matrix is due to the “liquid-like” chain relaxation in the MAF. This T_g (MAF) peak also shows a depression behavior for blends containing either IL-C10 or IL-108. But it is more unclear to see the T_g (MAF) peaks when the amount of ILs in blends increased up to 10 wt

Table 2. Temperature Values of Amorphous Relaxation Peaks by DEA and DMA

sample	T_g (MAF)			T_g (RAF)	
	DEA ^a (°C)	DEA ^b (°C)	DMA ^c (°C)	DEA ^b (°C)	DMA ^c (°C)
P(VDF-CTFE)-neat	−28	−31	−45	56	NA
P(VDF-CTFE)/IL-C18-2 wt %	−32	−33	−47	50	NA
P(VDF-CTFE)/IL-C18-5 wt %			−53		21
P(VDF-CTFE)/IL-C18-10 wt %	−44	−44	NA	28	18
P(VDF-CTFE)/IL-108-2 wt %	−36	−35	−47	48	NA
P(VDF-CTFE)/IL-108-5 wt %			NA		12
P(VDF-CTFE)/IL-108-10 wt %	−62	NA	NA	9	6

^aDielectric $\tan \delta$ vs T (Figure 9A). ^bDielectric M'' vs T (Figure 9B). ^cDynamic mechanical $\tan \delta$ vs T (Figure 10).

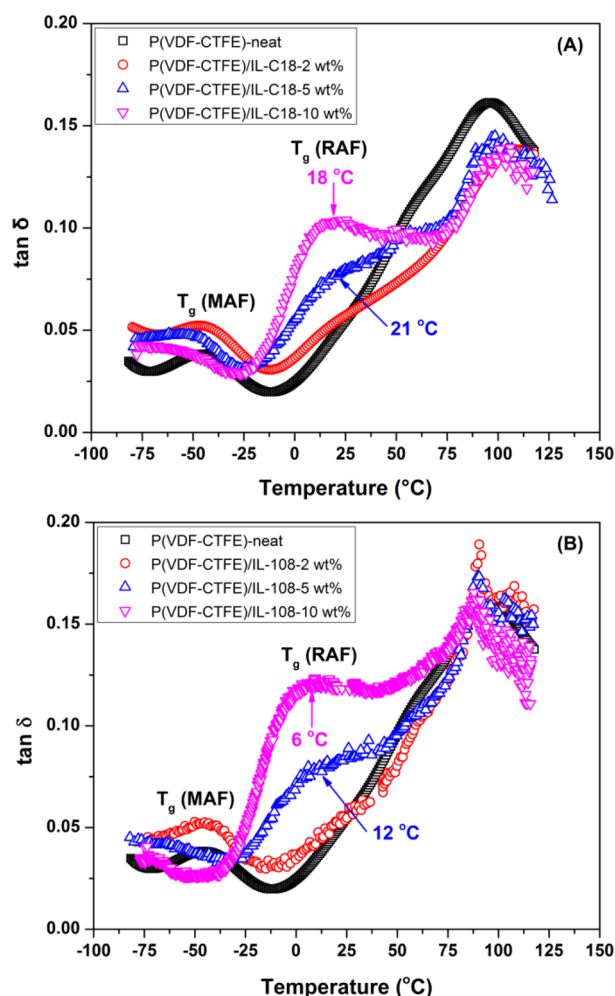


Figure 10. Dynamic mechanical loss tangent ($\tan \delta$) as a function of temperature for neat P(VDF-CTFE) and all blends with IL-C18 (A) and IL-108 (B). Note: the relaxation peaks from 75 to 100 °C are ascribed to imperfections in the crystalline phase,^{23,56} which can also be verified by the presence in the first heating run (Figure S1A) and absence in the second heating run (Figure S1C) from the DSC thermograms in the Supporting Information.

% for IL-C18 and 5 wt % for IL-108. It could be ascribed to the more miscible nature of the mixture of ILs and P(VDF-CTFE) chains in the MAF through the strong dipolar interaction which is more tardy to the dynamic changes of stress. The second T_g (RAF) relaxation peaks are not shown for neat P(VDF-CTFE) and blends containing 2 wt % of both ILs because of the very weak intensity in dynamic mechanical spectroscopy.²³ With gradually adding both ILs from 5 to 10 wt %, the RAF relaxation peaks are more visible and shift to lower temperature. Moreover, IL-108 induces lower values of T_g (RAF) than IL-C18 with the same IL amount. The depression behaviors of two T_g s corresponding to relaxations in the MAF and RAF of P(VDF-CTFE) are consistent with the DEA results.

The T_g (RAF) depression behavior unambiguously proves the localization of both phosphonium ILs in the RAF. On the one hand, as already mentioned above, the crystalline lamellae of P(VDF-CTFE) is the “template” that guides those ILs diffusing into the RAF constrained within the regular 2D crystalline lamella, inducing orderly assembled ILs with 1D to 2D geometric evolution with increasing the IL amount. On the other hand, the regular 1D or 2D assembled structure of ILs in

turn works as a so-called “template” to affect the polymer chain-folding to stabilize more *trans*-configuration, consequently inducing more polar γ - and/or β -crystalline phase. Actually, these regularly assembled structures of ILs in the RAF can be seen as those long tangible templates which can provide a long arrangement of surface functional groups that anchor the interaction with PVDF chains, such as IL-modified CNTs,^{25,26} OMMT,⁴⁰ and mesoporous SiO₂ nanorods.⁴¹ Meanwhile, these so-called “templates” in the RAF are also like obstacles to depress T_c , T_m , crystallinity, and crystallite size during crystallization through dipolar interactions with P(VDF-CTFE). Thus, the synergistic influence of mutual “templates” above synchronously occurring upon crystallization of film formation from solution endues the ILs efficient and versatile nanostructuration ability on the semicrystalline fluorinated matrix. In addition, the differences in the diffusion and interacting abilities of ILs due to the chemical structure govern the localization state of ILs in the RAF and thus tailor the nanostructuration on the semicrystalline fluorinated matrix. IL-108 with faster migration rate and stronger affinity with P(VDF-CTFE) is sure to be located in the RAF more than IL-C18 with larger distance between lamellae (confirmed by increased L_p values with the same IL amount) and thus induces and stabilizes more *trans*-sequence (β -phase) and also depresses the crystallization in a higher degree.

4. CONCLUSIONS

In conclusion, phosphonium ILs were found to be versatile and tunable nanostructuration agents on a semicrystalline fluorinated copolymer P(VDF-CTFE), including crystalline phase structure, dispersion morphology, and crystallization behavior. A fundamental mechanism of nanostructuration was elaborated in-depth. We proposed that the nanostructuration effect is strongly dependent on the diffusion abilities of ILs and the dipolar interaction strengths of ILs with P(VDF-CTFE) due to the difference in the chemical structures. Through the dipolar interaction, those ILs migrating into the RAF are constrained as a regular assembly templated by the 2D crystalline lamella. Simultaneously, this regular arrangement of ILs coupling with the strong dipolar interaction with P(VDF-CTFE) chains in the RAF in turn affects the chain folding to induce a more *trans*-sequence for polar crystalline phases and also restrains the crystallization to reduce T_c , T_m , crystallinity, and crystallite size. The synergistic effect of these mutual “templates” makes it possible for the nanostructuration of ILs. Since the diffusion ability and the dipolar interaction of IL with polymer matrix can be finely tuned by changing the chemical structures of cation and/or anion, the nanostructuration can also be easily tuned to require different applications. IL-108 shows a higher extent to be located in the RAF as well as stronger interaction with polymer matrix than IL-C18 due to the smaller steric hindrance and extra dipolar groups in the chemical structure, consequently inducing more polar β -phase and more homogeneous dispersion morphology and depressing the crystallization much more. Therefore, tailoring the chemical structures of phosphonium ILs based on the varied cation/anion combinations and small amount (from 2 wt %) in the PVDF-based matrix provides a facile and efficient pathway to adjust the profiles of blends through the tunable nanostructuration of ILs, which is of great potentiality for diverse applications of pyro/ferro/piezo-electric PVDF-based materials.

■ ASSOCIATED CONTENT

● Supporting Information

All DSC thermograms (Figure S1 and Table S1) and stress-strain tests results (Table S2). The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.5b00931.

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Notes

The authors declare no competing financial interest.

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